

(1) Murphy 3.24

Formaldehyde (HCHO) is produced by partial oxidation of methanol (CH₃OH). Several side reactions also occur, producing formic acid (HCOOH), CO, CO₂, and H₂O. Suppose 100 kgmol/h methanol and 21.05 kgmol/h O₂ are fed to a reactor, where 40% of the methanol and 95% of the O₂ is converted to products. What must the reactor outlet composition be?

Feed

100 kgmol/h methanol, CH₃OH

21.05 kgmol/h oxygen, O₂

Given

40% of the methanol is converted

95% of the oxygen is converted

Products include

Formaldehyde HCHO

Formic acid HCOOH

CO

CO₂

H₂O

Plus unreacted CH₃OH and O₂

$N_{\text{CH}_3\text{OH},\text{out}} = 100(1-0.40) = 60 \text{ kgmol/h}$

$100-60 = 40 \text{ kgmol/h (methanol reacted)}$

Oxygen

$N_{\text{O}_2,\text{out}} = 21.05(1-0.95) = 1.0525 \text{ kgmol/h}$

$21.05 - 1.0525 = 19.9975 = \text{basically just } 20 \text{ kgmol/h (} 20.0025 \text{)}$

N_{HCHO}

N_{HCOOH}

N_{CO}

N_{CO_2}

N_{H_2O}

5 unknowns

C balance

H balance

O balance

3 equations

DOF = 2

The outlet composition cannot be determined uniquely from the information given

What must be true is only:

$CH_3OH = 60 \text{ kgmol/h}$

$O_2 = 1.0525 \text{ kgmol/h}$

The problem is underdetermined by 2 degrees of freedom. I don't think I can solve for this without something about the side reactions or product distribution

should set all extents
except 1 to 0,
then solve
- 10 pts
15/25

(2) At low to moderate pressures, the equilibrium state of the water-gas shift reaction

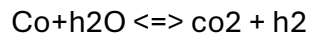
$$\text{CO} + \text{H}_2\text{O} \leftrightarrow \text{CO}_2 + \text{H}_2$$

Is approximately described by the relation

$$\frac{x_{\text{CO}_2} x_{\text{H}_2}}{x_{\text{CO}} x_{\text{H}_2\text{O}}} = K_e(T) = a \exp\{b/T\}; a = 0.0247 \text{ and } b = 4020 \text{ K}$$

where T is the reactor temperature in Kelvin, K_e is the reaction equilibrium constant, and x_i are the mole fractions of species i in the reactor at equilibrium. The feed to the reactor contains 20 mole % CO, 10 mole % CO₂, 40 mole % water, and the balance is inert gas. The reactor is maintained at $T = 1123 \text{ K}$.

- Assume a basis of 1 mole of feed and draw and label a flowchart. Carry out a degree-of-freedom analysis of the reactor based on the extent of reaction.
- Calculate the total moles of gas in the reactor at equilibrium and then the equilibrium mole fraction of hydrogen in the product.
- Suppose a gas sample is drawn from the reactor and analyzed shortly after startup and the mole fraction of hydrogen is significantly different from the calculated value. Assuming the calculations are correct, what is a likely explanation for the discrepancy?
- Use a method numerical method of your choice to take the input of the reactor temperature and the input feed component mole fractions of CO, H₂O, and CO₂ (assume no hydrogen is fed) and to calculate the mole fraction x_{H_2} in the product gas when equilibrium is reached. Present your results in a table. Use your numerical method to try and maximize the yield of hydrogen by adjusting the temperature and feed composition.



$$(X_{\text{co}_2} \cdot x_{\text{h}_2}) / (x_{\text{co}} \cdot x_{\text{h}_2\text{o}}) = K_c(T) = a \cdot \exp(b/t)$$

$$A = 0.0247$$

$$B = 4020 \text{ K}$$

$$T = 1123 \text{ K}$$

Feed composition

20 mol% CO

10 mol% CO₂

40 mol% H₂O

30 mol% inert

Basis of 1 mol feed

Diagram?
 ✓ 2 pts

Feed moles

$$\text{CO} = 0.2$$

$$\text{CO}_2 = .1$$

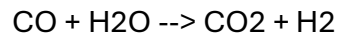
$$\text{H}_2\text{O} = 0.40$$

$$\text{Inert} = 0.30$$

$$\text{H}_2 = 0$$

Extent of reaction is "Xi"

Equilibrium moles



$$N_{\text{CO}} = 0.20 - x_i$$

$$N_{\text{H}_2\text{O}} = 0.40 - x_i$$

$$N_{\text{CO}_2} = 0.10 + x_i$$

$$N_{\text{H}_2} = x_i$$

$$N_i = 0.30$$

Unknown is x_i

One independent equation

$$\text{So DOF} = 0$$

2b

Total moles and equilibrium mole fraction of hydrogen

Since total moles remain constant:

$$N_t = 1.00 \text{ mol}$$

$$K_c = 0.0247 * \exp(4020/1123)$$

$$4020/1123 = 3.58$$

$$e^{3.58} = 35.9$$

$$K_c = 0.0247 * 35.8 = 0.886$$

$$(x_{\text{CO}_2} * x_{\text{H}_2}) / (x_{\text{CO}} * x_{\text{H}_2\text{O}}) = ((0.10 + x_i) * (x_i)) / ((0.20 - x_i)(0.40 - x_i)) = 0.886$$

$$(0.10 + x_i) * x_i = 0.886 * (0.20 - x_i)(0.40 - x_i)$$

$$0.10x_i + x_i^2 = 0.07088 - 0.5316x_i + 0.886x_i^2$$

$$0.114x_i^2 + 0.6316x_i - 0.07088 = 0$$

$$x_i = 0.110$$

$$x_{h2} = n_{h2}/n_t = x_i/1 = 0.110$$

$$n_t = 1.00 \text{ mol}$$

$$x_{h2} = 0.110$$

2c why might a startup sample differ from the equilibrium value?

The sample is taken shortly after startup

This is most likely because the reactor contents have not yet reached equilibrium.

2d

Numerical method and maximizing hydrogen yield

For a general feed basis of 1 mol with:

$$x_{co,0}$$

$$x_{h2o,0}$$

$$x_{co2,0}$$

$$Inert = 1 - x_{co,0} - x_{h2o,0} - x_{co2,0}$$

And no hydrogen initially, equilibrium gives

$$CO = x_{co,0} - x_i$$

$$H_2O = x_{h2o,0} - x_i$$

$$CO_2 = x_{co2,0} + x_i$$

$$H_2 = x_i$$

$$((x_{co2,0} + x_i)x_i) / ((x_{co,0} - x_i)(x_{h2o,0} - x_i)) = K(T)$$

$$K(T) = 0.0247 \exp(4020/T)$$

$$\exp(x) = e^x \text{ i just hate using } ^$$

$$T = 1123K$$

23/25

$$N_{co,0} = 0.20, n_{co2,0} = 0.10, n_{h2o,0} = 0.40, n_{h2,0} = 0$$

$$N_{co} = 0.20 - x_i$$

$$N_{h2o} = 0.40 - x_i$$

$$N_{co2} = 0.10 + x_i$$

$$N_{h2} = x_i$$

$$N_t = 1$$

$$\frac{((0.10+x_i)(x_i))}{((0.20-x_i)(0.40-x_i))} = K_c(T)$$

$$F(x_i) = \frac{((0.10+x_i)(x_i))}{((0.20-x_i)(0.40-x_i))} - K_c(T)$$

$$F(x_i) = 0$$

$$X_{h2} = x_i$$

Using bisection method

$$0 \leq x_i \leq \min(0.20, 0.40) = 0.20$$

Initial bracket is [0,0.19]

$$X_i = 0$$

$$F(0) = \frac{((0.10)(0))}{((0.20)(0.40))} - 0.8858 = -0.8858$$

$$X_i = 0.19$$

$$F(0.19) = \frac{((0.29)(0.19))}{((0.01)(0.21))} - 0.8858 = 25.3523$$

$F(0) < 0, f(0.19) > 0$ so there is a root between these brackets

Iteration	xL	xU	xm	f(xm)	New interval
1	0.000000	0.190000	0.095000	-0.307354	([0.095000, \ 0.190000])
2	0.095000	0.190000	0.142500	1.448088	([0.095000, \ 0.142500])
3	0.095000	0.142500	0.118750	0.250944	([0.095000, \ 0.118750])
4	0.095000	0.118750	0.106875	-0.075846	([0.106875, \ 0.118750])
5	0.106875	0.118750	0.112813	0.073006	([0.106875, \ 0.112813])
6	0.106875	0.112813	0.109844	-0.004671	([0.109844, \ 0.112813])
7	0.109844	0.112813	0.111328	0.033311	([0.109844, \ 0.111328])
8	0.109844	0.111328	0.110586	0.014112	([0.109844, \ 0.110586])
9	0.109844	0.110586	0.110215	0.004669	([0.109844, \ 0.110215])
10	0.109844	0.110215	0.110029	-0.000014	([0.110029, \ 0.110215])
11	0.110029	0.110215	0.110122	0.002324	([0.110029, \ 0.110122])

12	0.110029	0.110122	0.110076	0.001155	([0.110029,\ 0.110076])
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After 12 iterations, the root is between $0.110029 \leq x_i \leq 0.110076$

Taking the midpoint as the estimate

$$X_i = (0.110029 + 0.110076)/2 = 0.11005$$

$$X_{H_2} = x_i$$

$$X_{H_2} \approx 0.110$$

Use the same numerical method to try to maximize hydrogen yield

$$F(x_i) = ((z_{CO_2} + x_i)(x_i))/((z_{CO} - x_i)(z_{H_2O} - x_i)) - 0.0247 \exp(4020/T) = 0$$

Z = feed mole fraction

Case	T(K)	Z CO	Z H ₂ O	Z CO ₂	Inert	K _c (T)	Numerical root $x_i = x_{H_2}$
Base case	1123	0.20	0.40	0.10	0.30	0.8858	0.1100
1	1000	0.20	0.40	0.10	0.30	1.3758	0.1253
2	900	0.20	0.40	0.10	0.30	2.1505	0.1400
3	1123	0.30	0.50	0.00	0.20	0.8858	0.1822
4	1000	0.30	0.50	0.00	0.20	1.3758	0.2013
5	900	0.30	0.50	0.00	0.20	2.1505	0.2198
6	1123	0.40	0.50	0.00	0.10	0.8858	0.2156
7	1000	0.40	0.50	0.00	0.10	1.3758	0.2397
8	900	0.40	0.50	0.00	0.10	2.1505	0.2635
9	1123	0.50	0.50	0.00	0.00	0.8858	0.2424
10	1000	0.50	0.50	0.00	0.00	1.3758	0.2699
11	900	0.50	0.50	0.00	0.00	2.1505	0.2973

Lower temperature increases hydrogen yield

$$\text{Because } K_c(T) = 0.0247 \exp(4020/T)$$

Decreasing T increases K_c , which shifts equilibrium toward products in this model

Removing CO_2 from the feed increases hydrogen yield

CO_2 is a product so starting with less product favors more forward reaction

Reducing inert increases hydrogen yield

This lets more of the feed be reactive species

The best tested case was

$T = 900 \text{ K}$, $Z_{\text{co}} = 0.50$, $z_{\text{h2o}} = 0.50$, $z_{\text{co2}} = 0$

Which gave

$X_{\text{h2}} = 0.2973$

(3) Murphy 4.10

Your company is interested in building a plant to make ibuprofen, using the new catalytic reaction scheme described in example 4.1 your job is to measure the reaction rate with different catalysts as a first step in designing a full scale reactor. In one experiment, you mix 134g isobutyl benzene (IBB, $\text{C}_{10}\text{H}_{14}$) with 134g acetic anhydride (AAn, $\text{C}_4\text{H}_6\text{O}_3$) in a laboratory-scale batch reactor, adjust the temperature, add some catalyst, and wait 1 h. at the end of the hour you stop the reaction, collect the material in the pot and send it for chemical analysis. The chemist reports back that the pot contains 27g IBB, 52g acetic anhydride, 121 g isobutyl acetophenone (IBA, $\text{C}_{12}\text{H}_{16}\text{O}$), and 68g acetic acid (AAc, CH_3COOH). When you see these results, you tell the chemist that the analysis is wrong. He protests. Who is right?

IBB, $\text{C}_{10}\text{H}_{14}$

$\text{MW} = 10(12) + 14(1) = 134 \text{ g/mol}$

AAn, $\text{C}_4\text{H}_6\text{O}_3$

$\text{MW} = 4(12) + 6(1) + 3(16) = 102 \text{ g/mol}$

IBA, $\text{C}_{12}\text{H}_{16}\text{O}$

$\text{MW} = 12(12) + 16(1) + 16 = 176 \text{ g/mol}$

Acetic acid, $\text{CH}_3\text{COOH} = \text{C}_2\text{H}_4\text{O}_2$

$\text{MW} = 2(12) + 4(1) + 2(16) = 60 \text{ g/mol}$

Initial moles

IBB:

$N_{\text{ibb},0} = 134/134 = 1 \text{ mol}$

AAn:

$$N_{aan,0} = 134/102 = 1.3137 \text{ mol}$$

Final moles

$$IBB: n_{ibb} = 27/134 = 0.2015 \text{ mol}$$

AAn:

$$N_{aan} = 52/102 = 0.5098 \text{ mol}$$

IBA:

$$n_{iba} = 121/176 = 0.6875 \text{ mol}$$

AAc:

$$n_{aac} = 68/60 = 1.1333 \text{ mol}$$

Carbon balance

Initial carbon moles

From IBB :

$$10(1) = 10$$

From AAn:

$$4(1.3137) = 5.2548$$

Total in:

$$C_{in} = 10 + 5.2548 = 15.2548$$

Final carbon moles

From IBB:

$$10(0.2015) = 2.015$$

From AAN:

$$4(0.5098) = 2.0392$$

From IBA:

$$12(0.6875) = 8.250$$

From AAc:

$$2(1.1333) = 2.2666$$

Total out

$$C_{out} = 2.015 + 2.0392 + 8.250 + 2.2666 = 14.5708$$

These are not equal

$$15.2548 \neq 14.5708$$

$$15.2548 - 14.5708 = 0.6840 \text{ mol C atoms missing}$$

Oxygen balance

Initial oxygen moles

Only AAn contains oxygen

$$O_{in} = 3(1.3137) = 3.9411$$

Final oxygen moles

From AAn:

$$3(0.5098) = 1.5294$$

From IBA:

$$1(0.6875) = 0.6875$$

From AAc:

$$2(1.1333) = 2.2666$$

Total:

$$O_{out} = 1.5294 + 0.6875 + 2.2666 = 4.4835$$

$$\text{Initial } 3.9411 \neq 4.4835 \text{ output}$$

Oxygen has appeared:

$$4.4835 - 3.9411 = 0.5424 \text{ mol O atoms}$$

Impossible

Total mass

$$134 + 134 = 268 \text{ g in}$$

$$27 + 52 + 121 + 68 = 268 \text{ g out}$$

29/25

So i am right, the chemist's reported analysis is wrong



(4) Murphy 4.19 Solve this problem using an atomic balance and using a species balance.

4.19 Waste streams from various processes in a complex chemical facility are combined and burned for heat, because the cost of purifying and recycling the components is prohibitive. On a typical day, the combined waste stream contains 2 mol% acetone $(\text{CH}_3)_2\text{CO}$, 14 mol% acetic anhydride $(\text{CH}_3\text{CO})_2\text{O}$, 36 mol% methane CH_4 , 20 mol% ethane C_2H_6 , 13 mol% N_2 , 4 mol% O_2 , 8 mol% CO , and 3 mol% H_2S . The waste stream is fed to a furnace along with 20% excess air and combusted. (for combustion reactions, excess air is calculated on the assumption that the only products of combustion are CO_2 , H_2O , N_2 , and SO_2 .) Analysis of the exit flue gas gives a CO_2/CO mass ratio of 4:1. The only other components in the flue gas are N_2 , O_2 , H_2O , and SO_2 . Calculate the composition of the exit flue gas. Assume air is 21 mol% O_2 and 79 mol% N_2 .

For the atomic balance, do a DoF analysis. For 4b, do not count any chemical reactions. For 5d, only count independent atomic balances.

For the species balance, you will need to write chemical reaction equations. For the hydrocarbon fuels, assume that these are oxidized to CO and H_2O , that is, partial combustion. Add one equation to oxidize CO to CO_2 to account for any complete combustion for all the hydrocarbons oxidation reactions. Do a DoF analysis to show the problem is well posed.

For my own sake, im writing page 105 here so i can put away the book

DOF analysis

1. Draw a flow diagram and choose a system
2. Choose components
3. Label the flow diagram to show the components in each stream and any accumulation in the system. Write down any chemical reactions
4. Count the number of independent variables
 - 4a. Coun the number of independent stream variables (the number of components in each stream)

- 4b. Count the number of independent system variables (the number of independent chemical reactions, plus the number of components that accumulate or are depleted in the system)
5. Count the number of independent equations:
- 5a. Count the number of specified flows. (usually there is only one flow specified, that serves as the basis)
- 5b. Count the number of stream composition specifications (if there are N components in a stream, there are at most N – 1 specified stream compositions. For a splitter, add (N-1)*(Ns-1), to the count, where Ns is the number of output streams.)
- 5c. Count the number of system performance specifications
- 5d. Count the number of material balance equations. (if there are N components in a system, then there are N material balance equations)

4a) atomic balance solution

100 mol waste stream is the basis

Feed comp is

Acetone = 2

Acetic anhydride = 14

Methane = 36

Ethane = 20

N₂ = 13

O₂ = 4

CO = 8

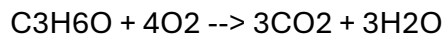
H₂S = 3

Computing stoichiometric oxygen for complete combustion

Excess air is based on the assumption of complete combustion to CO₂, H₂O, N₂, SO₂

Acetone: C₃H₆O

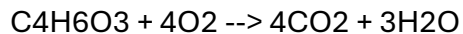
For complete combustion:



For 2 mol:

$$2(4) = 8 \text{ mol } O_2$$

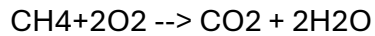
Acetic anhydride: C₄H₆O₃



For 14 mol:

$$14(4) = 56$$

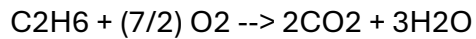
Methane



For 36 mol:

$$36(2) = 72$$

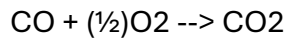
Ethane



For 20 mol:

$$20(3.5) = 70$$

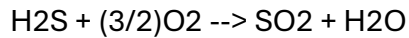
CO



For 8 mol:

$$8(0.5) = 4$$

H₂S



For 3 mol:

$$3(1.5) = 4.5$$

Total O₂ needed:

$$8 + 56 + 72 + 70 + 4 + 4.5 = 214.5$$

But 4 mol O₂ already enters in the waste stream so

$$214.5 - 4 = 210.5 \text{ mol O}_2$$

20% excess air means

$$\text{O}_{2,\text{air}} = 1.20(210.5) = 252.6 \text{ mol}$$

Air is 21% O₂, 79% N₂, so nitrogen entering with air is

$$\text{N}_{2,\text{air}} = 252.6(79/21) = 950.9714$$

Total nitrogen entering:

$$\text{N}_{2,\text{in}} = 13 + 950.9714 = 963.9714$$

Define product amounts

n_{CO_2} , n_{CO} , $n_{\text{H}_2\text{O}}$, n_{SO_2} , n_{NO_2} , n_{N_2}

Using the given CO₂/CO mass ratio

$$M_{\text{CO}_2}/M_{\text{CO}} = 4$$

$$44n_{\text{CO}_2}/28n_{\text{CO}} = 4$$

$$44n_{\text{CO}_2} = 112n_{\text{CO}}$$

$$n_{\text{CO}_2} = 28/11 * n_{\text{CO}}$$

Carbon atomic balance

$$\text{Acetone: } 2(3) = 6$$

$$\text{Acetic anhydride: } 14(4) = 56$$

$$\text{Methane: } 36(1) = 36$$

Ethane: $20(2) = 40$

CO: $8(1) = 8$

Total carbon in:

$$6+56+36+40+8 = 146$$

All carbon leaves as CO_2 or CO

$$n_{\text{CO}_2} + n_{\text{CO}} = 146$$

$$n_{\text{CO}_2} = 28/11 * n_{\text{CO}}$$

$$28/11 * n_{\text{CO}} + n_{\text{CO}} = 146$$

$$n_{\text{CO}} = 41.1795$$

$$n_{\text{CO}_2} = 146 - 41.1795 = 104.8205$$

Hydrogen atomic balance

$$\text{Acetone: } 2(6) = 12$$

$$\text{Acetic anhydride: } 14(6) = 84$$

$$\text{Methane: } 36(4) = 144$$

$$\text{Ethane: } 20(6) = 120$$

$$\text{H}_2\text{S: } 3(2) = 6$$

Total:

$$12+84+144+120+6 = 366$$

All hydrogen leaves as water:

$$2n_{\text{H}_2\text{O}} = 366$$

$$n_{\text{H}_2\text{O}} = 183$$

Sulfur atomic balance

All sulfur leaves as SO_2 :

$$n_{\text{SO}_2} = 3$$

Nitrogen

$$n_{\text{N}_2} = 963.9714$$

Oxygen atomic balance

Oxygen in:

$$\text{Acetone: } 2(1) = 2$$

$$\text{Acetic anhydride: } 14(3) = 42$$

$$\text{Feed oxygen gas: } 4(2) = 8$$

$$\text{Feed CO: } 8(1) = 8$$

$$\text{Air oxygen gas: } 252.6(2) = 505.2$$

Total oxygen atoms in:

$$2 + 42 + 8 + 8 + 505.2 = 565.2$$

Oxygen atoms out in known products:

$$\text{CO}_2: 2(104.8205) = 209.6410$$

$$\text{CO: } 41.1795$$

H₂O: 183

SO₂: 2(3) = 6

O₂: 2N_{o2}

$$565.2 = 209.6410 + 41.1795 + 183 + 6 + 2n_{O_2}$$

$$565.2 = 439.8205 + 2n_{O_2}$$

$$2n_{O_2} = 125.3795$$

$$N_{O_2} = 62.6898$$

Total flue gas and composition

$$N_t = 104.8205 + 41.1795 + 183 + 3 + 62.6898 + 963.9714 = 1358.6612$$

Mole fractions

$$Y_{CO_2} = 104.8205/1358.6612 = 0.07715$$

$$Y_{CO} = 41.1795/1358.6612 = 0.03031$$

$$Y_{H_2O} = 183/1358.6612 = 0.13469$$

$$Y_{SO_2} = 3/1358.6612 = 0.00221$$

$$Y_{O_2} = 62.6898/1358.6612 = 0.04614$$

$$Y_{N_2} = 963.9714/1358.6612 = 0.70950$$

Atomic balance answer

$$CO_2 = 104.82 \text{ mol}$$

$$CO = 41.18 \text{ mol}$$

$$H_2O = 183 \text{ mol}$$

$$SO_2 = 3 \text{ mol}$$

$$O_2 = 62.69 \text{ mol}$$

$$N_2 = 963.97 \text{ mol}$$

✓

125/125

4B) DOF analysis

Waste feed + air --> furnace --> flue gas

Steady state, so no accumulation, splitter

4a 7

4b 0

7

5a 7

5b 1

5c 8

5c 2

5d 5

16

Reducing the dof

Unknowns:

Air flowrate A

N_{CO_2}

N_{CO}

N_{N_2}

N_{O_2}

N_{H_2O}

N_{SO_2}

7

Independent equations

Excess air specification: 1

CO_2/CO mass ratio specification: 1

Independent atomic balances C,H,O,N,S 5

Total = 7

$7 - 7 = 0$

Dof = 0